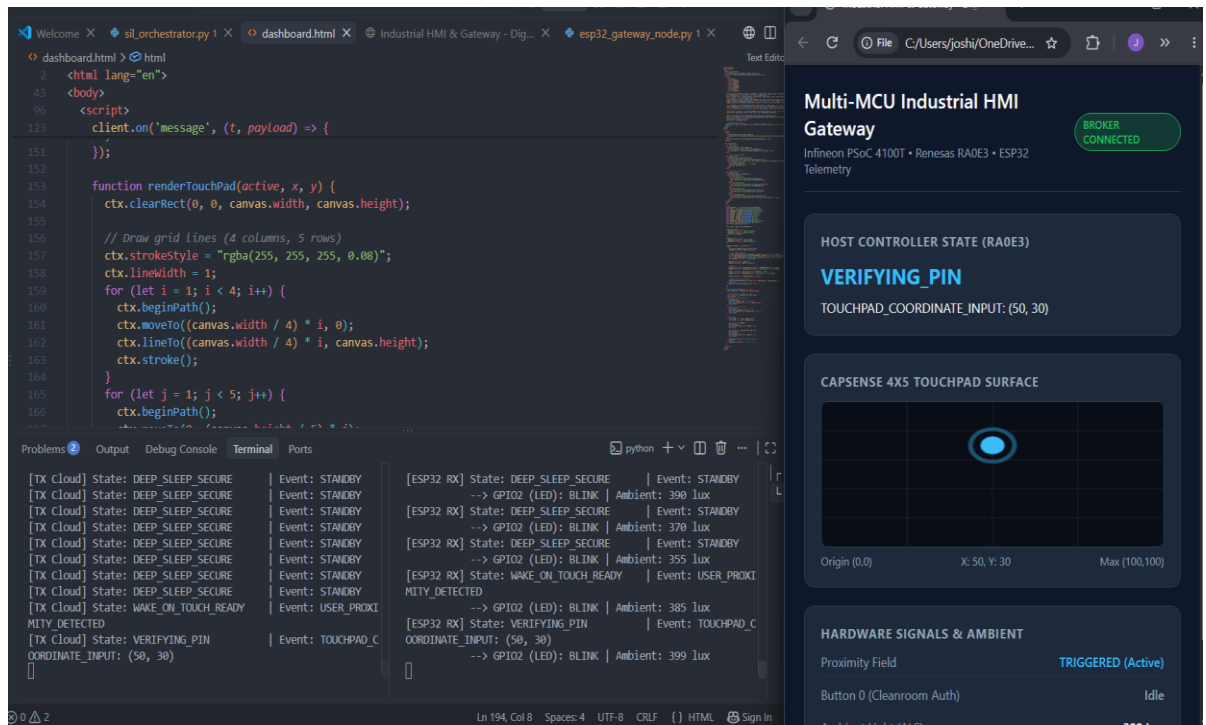
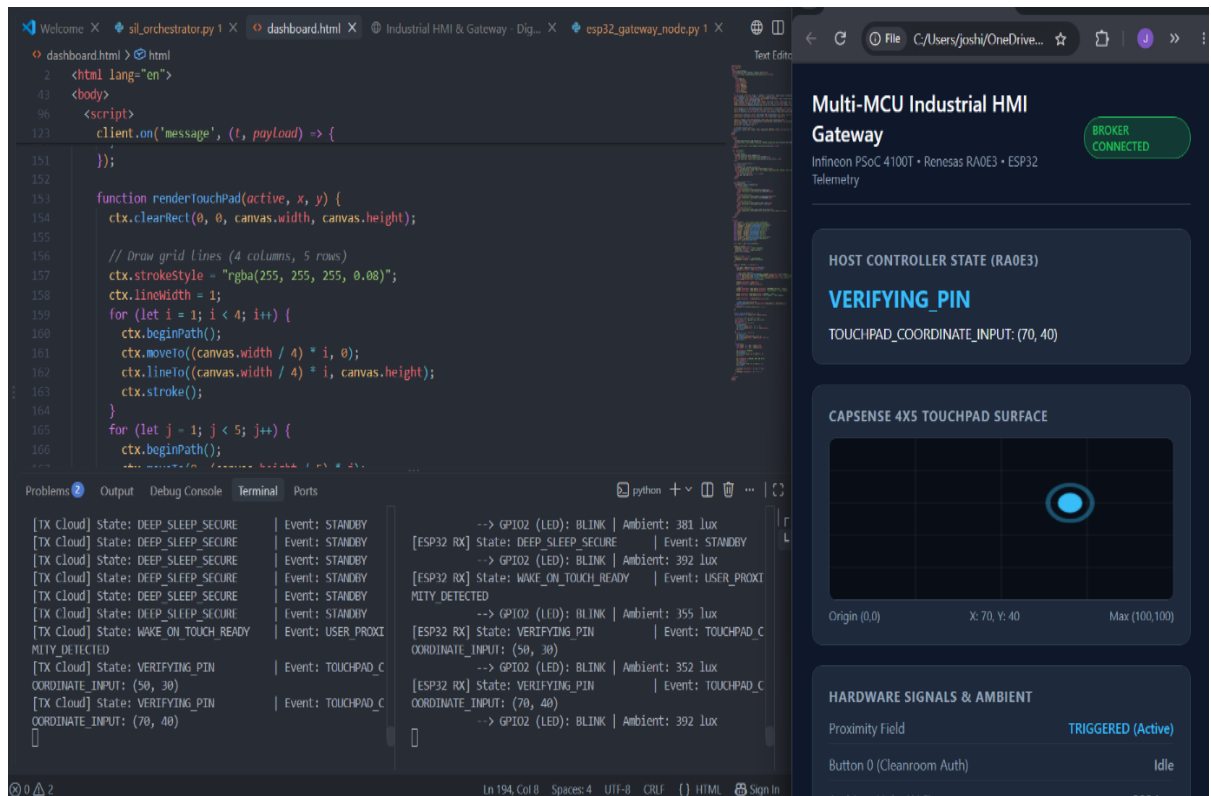


OUTPUT IMAGES:

1. Telemetry Verification: Coordinate (50, 30)



2. Telemetry Verification: Coordinate (70, 40) & Ambient ALS



3. Interlock Triggered: SYSTEM LOCKED.

```
sil_orchestrator.py 1 X virtual_psoc.py X C ra0e3_fsm.c 8 X ra0e3_host_node.py X dashboard.html X Industrial HMI & Gateway - Dig... X esp32_gateway_node.py 1 X
virtual_psoc.py > ...
46 def run_server():
84     except (BrokenPipeError, ConnectionResetError):
85         print("[PSoC 4100T] Host Controller disconnected.")
86     finally:
87         conn.close()
88         server.close()
89         print("[PSoC 4100T] Virtual sensor simulation terminated.")
90
91 if __name__ == "__main__":
92     run_server()

Problems 10 Output Debug Console Terminal Ports
inGW-w64
No package found matching input criteria.
PS C:\Users\joshi\OneDrive\Desktop\SELF PROJECT\VCU> python ra0e3_host_node.py

[Renesas RA0E3 Host Controller] Initializing SIL FSM Engine...

[PSoC 4100T] Connecting to PSoC 4100T Virtual I2C bus at 127.0.0.1:5000...
[PSoC 4100T] I2C Bus Link Active. Listening for sensor events...

[I2C Read PASS] Checksum Verified: 0xD0
-> Host State: PIN_ENTRY_MODE | Lux: 464 | Coords: (0, 0)

[I2C Read PASS] Checksum Verified: 0xF5
-> Host State: AUTH_FAILED_LOCKED | Lux: 486 | Coords: (48, 42)

[I2C Read PASS] Checksum Verified: 0xCE
-> Host State: AUTH_FAILED_LOCKED | Lux: 461 | Coords: (0, 0)

[Step 03] Injected Event Mode: BUTTON0
Status: 0x02 (BUTTON_0_PRESS) | ALS: 461 Lux
Raw I2C Bytes Sent: ['0x2', '0x0', '0x0', '0x1', '0xcd', '0xce']

[Step 04] Injected Event Mode: BUTTON1
Status: 0x04 (BUTTON_1_PRESS) | ALS: 464 Lux
Raw I2C Bytes Sent: ['0x4', '0x0', '0x0', '0x1', '0xd0', '0xd5']

[Step 05] Injected Event Mode: CORRUPT
Status: 0x00 (IDLE / NOISE) | ALS: 474 Lux
Raw I2C Bytes Sent: ['0x0', '0x0', '0x0', '0x1', '0xda', '0xda']

[Step 06] Injected Event Mode: TOUCHPAD
Status: 0x08 (TOUCHPAD_COORDINATES) | ALS: 478 Lux
Raw I2C Bytes Sent: ['0x8', '0x22', '0x3e', '0x1', '0xde', '0xcb']

[PSoC 4100T] Virtual sensor simulation terminated.
PS C:\Users\joshi\OneDrive\Desktop\SELF PROJECT\VCU>
```

4. [I2C Read FAIL] Checksum mismatch! Corrupted packet rejected

```
sil_orchestrator.py 1 X virtual_psoc.py X C ra0e3_fsm.c 8 X ra0e3_host_node.py X dashboard.html X Industrial HMI & Gateway - Dig... X esp32_gateway_node.py 1 X
virtual_psoc.py > ...
46 def run_server():
84     except (BrokenPipeError, ConnectionResetError):
85         print("[PSoC 4100T] Host Controller disconnected.")
86     finally:
87         conn.close()
88         server.close()
89         print("[PSoC 4100T] Virtual sensor simulation terminated.")
90
91 if __name__ == "__main__":
92     run_server()

Problems 10 Output Debug Console Terminal Ports

[I2C Read PASS] Checksum Verified: 0xD0
-> Host State: PIN_ENTRY_MODE | Lux: 464 | Coords: (0, 0)

[I2C Read PASS] Checksum Verified: 0xF5
-> Host State: AUTH_FAILED_LOCKED | Lux: 486 | Coords: (48, 42)

[I2C Read PASS] Checksum Verified: 0xCE
-> Host State: AUTH_FAILED_LOCKED | Lux: 461 | Coords: (0, 0)

[I2C Read PASS] Checksum Verified: 0xD5
-> Host State: AUTH_FAILED_LOCKED | Lux: 464 | Coords: (0, 0)

[I2C Read FAIL] Checksum mismatch! Corrupted packet rejected (0x24)

[I2C Read PASS] Checksum Verified: 0xCB
-> Host State: AUTH_FAILED_LOCKED | Lux: 478 | Coords: (34, 62)

[PSoC 4100T] Virtual I2C bus closed. Exiting.
PS C:\Users\joshi\OneDrive\Desktop\SELF PROJECT\VCU>

[Step 03] Injected Event Mode: BUTTON0
Status: 0x02 (BUTTON_0_PRESS) | ALS: 461 Lux
Raw I2C Bytes Sent: ['0x2', '0x0', '0x0', '0x1', '0xcd', '0xce']

[Step 04] Injected Event Mode: BUTTON1
Status: 0x04 (BUTTON_1_PRESS) | ALS: 464 Lux
Raw I2C Bytes Sent: ['0x4', '0x0', '0x0', '0x1', '0xd0', '0xd5']

[Step 05] Injected Event Mode: CORRUPT
Status: 0x00 (IDLE / NOISE) | ALS: 474 Lux
Raw I2C Bytes Sent: ['0x0', '0x0', '0x0', '0x1', '0xda', '0xda']

[Step 06] Injected Event Mode: TOUCHPAD
Status: 0x08 (TOUCHPAD_COORDINATES) | ALS: 478 Lux
Raw I2C Bytes Sent: ['0x8', '0x22', '0x3e', '0x1', '0xde', '0xcb']

[PSoC 4100T] Virtual sensor simulation terminated.
PS C:\Users\joshi\OneDrive\Desktop\SELF PROJECT\VCU>
```